

Homework 3

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Physics 16

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Question 1. Purcell chapter 1, exercise 1.60 (Field from a hollow cylinder.) [10 pts] Consider the hollow cylinder from Exercise 1.59 (previous problem set, PS4). Use Gauss's law to show that the field inside the pipe is zero. Also show that the field outside is the same as if the charge were all on the axis. Is either statement true for a pipe of square cross section on which the charge is distributed with uniform surface density?

Solution. Consider an infinitely long hollow circular cylinder of radius R carrying a uniform surface charge density σ , so the charge per unit length is $\lambda = 2\pi R\sigma$. Take a coaxial cylinder of radius r and length L as the Gaussian surface. By symmetry, $\vec{E}$ is perpendicular to the curved side and parallel to the end caps, so the end caps contribute nothing to the flux, hence

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = E(r) \cdot 2\pi rL.$$

Inside the pipe ($r < R$), no charge is enclosed, so Gauss's law gives

$$E(r) \cdot 2\pi rL = \frac{Q_{\text{enc}}}{\epsilon_0} = 0 \implies \vec{E} = 0 \quad \text{for } r < R.$$

Outside the pipe ($r > R$), the enclosed charge is $Q_{\text{enc}} = \lambda L$, so

$$E(r) \cdot 2\pi rL = \frac{\lambda L}{\epsilon_0} \implies \vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} \quad \text{for } r > R.$$

This is identical to the field of an infinite line of charge λ placed on the axis, so outside the pipe the field is the same as if all the charge were concentrated on the axis. If the pipe has a square cross section with uniform surface charge density on its four faces of length ℓ , the rotational symmetry is lost. Consequently,

- *Inside:* Although Gauss's law still gives zero total flux through any closed surface in the interior, this does not imply $\vec{E} = 0$. A point near one face is closer to that face than to the opposite one, so the fields of the four faces do not cancel, and $\vec{E} \neq 0$ in general.
- *Outside:* The field is *not* the same as that of a line charge on the axis. Only at distances $r \gg \ell$ does the field approach the line-charge form.

□

Question 2. Purcell chapter 2, exercise 2.37 (Field on Earth.) [5 pts] A sphere the size of the Earth has 1 C of charge distributed evenly over its surface. What is the electric field strength just outside the surface? What is the potential of the sphere, with zero potential at infinity?

Solution. For a uniformly charged sphere, the field and potential outside are identical to those of a point charge Q located at the center. Using the Earth's radius $R_E \approx 6.37 \times 10^6$ m and $Q = 1$ C, with $\frac{1}{4\pi\epsilon_0} \approx 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$,

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R_E^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1 \text{ C})}{(6.37 \times 10^6 \text{ m})^2} \approx \boxed{2.22 \times 10^{-4} \text{ N/C}},$$

and

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{Q}{R_E} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1 \text{ C})}{6.37 \times 10^6 \text{ m}} \approx \boxed{1.41 \times 10^3 \text{ V}}.$$

□

Question 3. Purcell chapter 2, exercise 2.72 (Electric field of spherical charge distribution, from given potential.) [10 pts] Find (and make rough plots of) the electric field and charge distribution that go with the potential given in the Purcell exercise text,

$$\phi = \begin{cases} \frac{\rho_0}{4\pi\epsilon_0}(x^2 + y^2 + z^2) & \text{for } x^2 + y^2 + z^2 < a^2 \\ \frac{\rho_0}{4\pi\epsilon_0} \left(-a^2 + \frac{2a^3}{(x^2 + y^2 + z^2)^{1/2}} \right) & \text{for } x^2 + y^2 + z^2 > a^2 \end{cases}$$

where ρ_0 is a quantity with the dimensions of volume charge density. Note that we are not assuming that $\phi = 0$ at infinity.

Solution. Since $x^2 + y^2 + z^2 = r^2$, the potential only depends on the radial coordinate, so we rewrite it in spherical coordinates:

$$\phi(r) = \begin{cases} \frac{\rho_0}{4\pi\epsilon_0} r^2 & r < a \\ \frac{\rho_0}{4\pi\epsilon_0} \left(-a^2 + \frac{2a^3}{r} \right) & r > a \end{cases}$$

First,

$$\vec{E} = -\nabla\phi = -\frac{\partial\phi}{\partial r} \hat{r}.$$

So,

$$\vec{E} = \begin{cases} -\frac{\rho_0 r}{2\pi\epsilon_0} \hat{r} & r < a \\ \frac{\rho_0 a^3}{2\pi\epsilon_0 r^2} \hat{r} & r > a \end{cases}$$

Evaluating both expressions at the boundary,

$$\vec{E}(a) = \begin{cases} -\frac{\rho_0 a}{2\pi\epsilon_0} \hat{r} & r \rightarrow a^- \\ +\frac{\rho_0 a}{2\pi\epsilon_0} \hat{r} & r \rightarrow a^+ \end{cases}$$

The directions flip, so E_r jumps by

$$\Delta E_r = E_{r,\text{out}} - E_{r,\text{in}} = \frac{\rho_0 a}{2\pi\epsilon_0} - \left(-\frac{\rho_0 a}{2\pi\epsilon_0} \right) = \frac{\rho_0 a}{\pi\epsilon_0}.$$

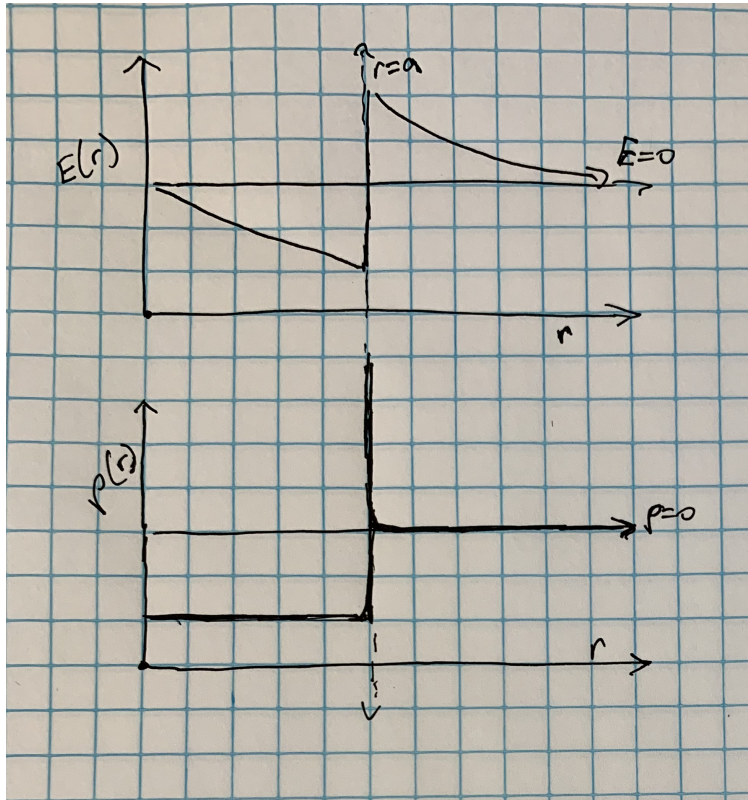
A discontinuity in the radial field implies a surface charge at $r = a$, with density

$$\sigma = \epsilon_0 \Delta E_r = \boxed{\frac{\rho_0 a}{\pi}}.$$

Now let us find charge density. Using $\rho = \epsilon_0(\nabla \cdot \vec{E}) = \frac{\epsilon_0}{r^2} \frac{\partial}{\partial r}(r^2 E_r)$,

$$\rho = \begin{cases} -\frac{3\rho_0}{2\pi} & r < a \\ 0 & r > a \end{cases}$$

For $r > a$, $E_r = \frac{\rho_0 a^3}{2\pi\epsilon_0 r^2}$, so $r^2 E_r = \frac{\rho_0 a^3}{2\pi\epsilon_0}$ is constant, and its derivative vanishes. We draw the graphs as such:



□

Question 4. Purcell chapter 2, exercise 2.73 (Laplace equation.) [5 pts] Does the function $f(x, y) = x^2 + y^2$ satisfy the two-dimensional Laplace equation? Does the function $g(x, y) = x^2 - y^2$? Sketch the latter function, calculate the gradient at the points $(x, y) = (0, 1)$, $(1, 0)$, $(0, -1)$, and $(-1, 0)$, and indicate by little arrows the directions in which these gradient vectors point.

Solution. The two-dimensional Laplacian is defined as the sum of the unmixed second partial derivatives:

$$\nabla^2 h(x, y) = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2}.$$

A function satisfies Laplace's equation when $\nabla^2 h = 0$. Testing the former equation gives

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2}(x^2 + y^2) + \frac{\partial^2 f}{\partial y^2}(x^2 + y^2) = 2 + 2 = 4 \neq 0.$$

Hence, f does *not* satisfy Laplace's equation. Now let us test the latter equation:

$$\nabla^2 g = \frac{\partial^2 f}{\partial x^2}(x^2 - y^2) + \frac{\partial^2 f}{\partial y^2}(x^2 - y^2) = 2 - 2 = 0.$$

Hence, g *does* satisfy Laplace's equation. The gradient of g is

$$\nabla g = \frac{\partial g}{\partial x} \hat{x} + \frac{\partial g}{\partial y} \hat{y} = \boxed{2x \hat{x} - 2y \hat{y}},$$

and

$$\nabla g(0, 1) = (0, -2)$$

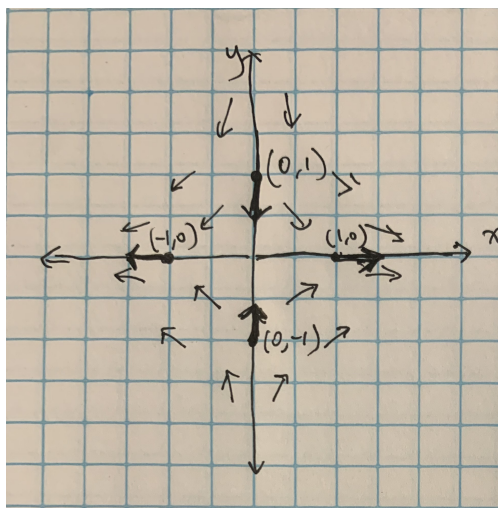
$$\nabla g(1, 0) = (2, 0)$$

$$\nabla g(0, -1) = (0, 2)$$

$$\nabla g(-1, 0) = (-2, 0).$$

We draw g as such:

□



Question 5. Purcell chapter 2, exercise 2.60 (Dipole.) [10 pts] Using the $r = r_0 \sin^2 \theta$ expression for the dipole $\vec{E}$ field lines in Purcell Fig. 2.19, find the locations where the curves are horizontal. (In other words, where does the electric field point perpendicularly to the dipole moment vector?)

Solution. Place the dipole moment $\vec{p}$ along the z -axis (the “vertical” direction), so θ is measured from $\hat{z}$. A horizontal field line is one whose tangent has no $\hat{z}$ -component, i.e. where $\vec{E} \perp \vec{p}$. In the plane containing $\vec{p}$, the cylindrical radius is $s = r \sin \theta$ and the vertical coordinate is $z = r \cos \theta$. Using $r = r_0 \sin^2 \theta$,

$$s = r_0 \sin^3 \theta, \quad z = r_0 \sin^2 \theta \cos \theta.$$

A horizontal tangent means $dz/ds = 0$, or equivalently $dz/d\theta = 0$ (since $ds/d\theta \neq 0$ where the tangent is horizontal). Hence,

$$\frac{dz}{d\theta} = r_0 (2 \sin \theta \cos^2 \theta - \sin^3 \theta) = r_0 \sin \theta (2 \cos^2 \theta - \sin^2 \theta).$$

Setting this to zero (and discarding $\sin \theta = 0$, which corresponds to points on the dipole axis itself),

$$2 \cos^2 \theta = \sin^2 \theta \implies \tan^2 \theta = 2 \implies \boxed{\theta = \pm \arctan \sqrt{2}.}$$

Hence, horizontal curves will exist in the cone formed when rotating a line of angle θ around the z -axis. □

Question 6. Purcell chapter 2, exercise 2.62 (Quadrupole field.) [15 pts] Consider a square quadrupole consisting of two adjacent dipoles oppositely oriented and placed side by side to form a square, as shown in Purcell Fig. 2.16. If the side length is ℓ , find the electric field at a large distance r along the diagonal containing the two positive charges. Be careful to take into account all quantities that are second order in ℓ/r .

Solution. We draw the problem as such: Notice that the force acted upon the two negative charges cancel out the direction perpendicular to the diagonal. Hence,

$$\begin{aligned} E_r &= \frac{kq}{(r-d)^2} + \frac{kq}{(r+d)^2} - \frac{2kq}{r^2+d^2} \cos \theta \\ &= \frac{kq}{(r-d)^2} + \frac{kq}{(r+d)^2} - \frac{2kqr}{(r^2+d^2)^{3/2}} \\ &= \frac{kq}{r^2} \left[\left(1 - \frac{d}{r}\right)^{-2} + \left(1 + \frac{d}{r}\right)^{-2} - 2 \left(1 + \frac{d^2}{r^2}\right)^{-3/2} \right]. \end{aligned}$$

Expanding each term to second order in d/r via $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$:

$$\begin{aligned} \left(1 - \frac{d}{r}\right)^{-2} &\approx 1 + 2\frac{d}{r} + 3\frac{d^2}{r^2}, \\ \left(1 + \frac{d}{r}\right)^{-2} &\approx 1 - 2\frac{d}{r} + 3\frac{d^2}{r^2}, \\ \left(1 + \frac{d^2}{r^2}\right)^{-3/2} &\approx 1 - \frac{3}{2}\frac{d^2}{r^2}. \end{aligned}$$

Adding the first two, the linear terms cancel and we get $2 + 6d^2/r^2$. Subtracting twice the third gives $2 - 3d^2/r^2$. Thus

$$E_r = \frac{kq}{r^2} \left[\left(2 + 6\frac{d^2}{r^2}\right) - \left(2 - 3\frac{d^2}{r^2}\right) \right] = \frac{9kq d^2}{r^4}.$$

Since the half-diagonal is $d = \ell/\sqrt{2}$, we have $d^2 = \ell^2/2$, so

$$\boxed{\vec{E} = \frac{9kq \ell^2}{2r^4} \hat{r} = \frac{9q \ell^2}{8\pi\epsilon_0 r^4} \hat{r}.}$$

□

Question 7. Purcell chapter 2, exercise 2.71 (A one-dimensional charge distribution.) [5 pts] Find (and make rough plots of) the electric field and charge distribution that go with the following potential: $\phi(x) = B(\ell^2 - x^2)$ for $|x| \leq \ell$, $\phi(x) = 0$ for $|x| > \ell$.

Solution. In one dimension, $\vec{E} = -\frac{d\phi}{dx} \hat{x}$ and $\rho = \varepsilon_0 \frac{dE_x}{dx}$ (from $\nabla \cdot \vec{E} = \rho/\varepsilon_0$). Differentiating ϕ gives the potential piecewise,

$$E_x = -\frac{d\phi}{dx} = \begin{cases} -\frac{d}{dx}[B(\ell^2 - x^2)] = 2Bx & |x| < \ell \\ -\frac{d}{dx}(0) = 0 & |x| > \ell \end{cases}$$

So

$$\vec{E}(x) = \begin{cases} 2Bx \hat{x} & |x| < \ell \\ 0 & |x| > \ell \end{cases}$$

At $x = +\ell$, the field inside is $E_x = 2B\ell$ and the field outside is $E_x = 0$, which is a discontinuity of magnitude $2B\ell$. The same thing happens at $x = -\ell$, where E_x jumps from $-2B\ell$ to 0, a jump of magnitude $2B\ell$. Now let us find ρ . For $|x| < \ell$,

$$\rho = \varepsilon_0 \frac{dE_x}{dx} = \varepsilon_0 \cdot 2B = 2\varepsilon_0 B.$$

For $|x| > \ell$, $E_x = 0$ so $\rho = 0$. Jumps in E_x signal sheet (surface) charges. Using $\sigma = \varepsilon_0(E_{x,\text{right}} - E_{x,\text{left}})$ at each boundary. At $x = +\ell$,

$$\sigma(\ell) = \lim_{h \rightarrow 0} \varepsilon_0(E_x(\ell + h) - E_x(\ell - h)) = \varepsilon_0(0 - 2B\ell) = -2\varepsilon_0 B\ell.$$

Similarly, at $x = -\ell$,

$$\sigma(-\ell) = \lim_{h \rightarrow 0} \varepsilon_0(E_x(-\ell + h) - E_x(-\ell - h)) = \varepsilon_0(-2B\ell - 0) = -2\varepsilon_0 B\ell.$$

Hence,

$$\rho(x) = \begin{cases} 2\varepsilon_0 B & |x| < \ell \\ 0 & |x| > \ell \end{cases}$$

and

$$\sigma = -2\varepsilon_0 B\ell, \quad x = \pm\ell.$$

We draw the graphs as such below:

□

